EXPERIMENT 1: Text Preprocessing using NLTK

AIM:

To perform text preprocessing on various corpora using NLTK library. The experiment demonstrates normalization, cleaning, tokenization, stopword removal, stemming, and lemmatization on Brown, Reuters, 20 Newsgroups, and UDHR datasets.

PROCEDURE:

1. Install required libraries: nltk, scikit-learn, pandas.
2. Download NLTK corpora: brown, reuters, stopwords, punkt, wordnet, omw-1.4.
3. Import required modules: re, nltk, brown, reuters, stopwords, word_tokenize, PorterStemmer, WordNetLemmatizer, fetch_20newsgroups.
4. Initialize stop_words, stemmer, and lemmatizer.
5. Define preprocess_text(text) that applies:
- Text Normalization (lowercase)
- Text Cleaning (remove punctuation and numbers using regex)
- Tokenization (word_tokenize)
- Stopword Removal
- Stemming (PorterStemmer)
- Lemmatization (WordNetLemmatizer)
6. Apply preprocessing to sample text from each corpus (Brown, Reuters, 20Newsgroups, UDHR).
7. Print tokens, stemmed, and lemmatized outputs for each.

CODE:

```
pip install nltk scikit-learn pandas

import nltk
nltk.download('brown')
nltk.download('reuters')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')

import re
import nltk
from nltk.corpus import brown, reuters, stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
from sklearn.datasets import fetch_20newsgroups
```

```
import pandas as pd

stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

def preprocess_text(text):
    # 1. Text Normalization (lowercase)
    text = text.lower()
    # 2. Text Cleaning (remove punctuation and numbers)
    text = re.sub(r'[^a-z\s]', '', text)
    # 3. Tokenization
    tokens = word_tokenize(text)
    # 4. Stopword Removal
    tokens = [word for word in tokens if word not in stop_words]
    # 5. Stemming
    stemmed = [stemmer.stem(word) for word in tokens]
    # 6. Lemmatization
    lemmatized = [lemmatizer.lemmatize(word) for word in tokens]
    return {"tokens": tokens, "stemmed": stemmed, "lemmatized": lemmatized}

# Brown corpus
brown_text = " ".join(brown.words()[:1000])
brown_processed = preprocess_text(brown_text)
print("Tokens:", brown_processed["tokens"][:20])
print("Stemmed:", brown_processed["stemmed"][:20])
print("Lemmatized:", brown_processed["lemmatized"][:20])

# Reuters corpus
reuters_text = " ".join(reuters.words()[:1000])
reuters_processed = preprocess_text(reuters_text)
print("Tokens:", reuters_processed["tokens"][:20])
print("Stemmed:", reuters_processed["stemmed"][:20])
print("Lemmatized:", reuters_processed["lemmatized"][:20])

# 20 Newsgroups
newsgroups = fetch_20newsgroups(subset='train')
newsgroup_text = newsgroups.data[0]
newsgroup_processed = preprocess_text(newsgroup_text)
print("Tokens:", newsgroup_processed["tokens"][:20])
print("Stemmed:", newsgroup_processed["stemmed"][:20])
print("Lemmatized:", newsgroup_processed["lemmatized"][:20])

# UDHR corpus
from nltk.corpus import udhr
nltk.download('udhr')
ud_text = " ".join(udhr.words('English-Latin1')[:1000])
ud_processed = preprocess_text(ud_text)
print("Tokens:", ud_processed["tokens"][:20])
print("Stemmed:", ud_processed["stemmed"][:20])
print("Lemmatized:", ud_processed["lemmatized"][:20])
```

OUTPUT:

```
Tokens: ['fulton', 'county', 'grand', 'jury', 'said', 'friday', 'investigation',
'atlanta', 'recent', 'primary', 'election', ...]
Stemmed: ['fulton', 'counti', 'grand', 'juri', 'said', 'friday', 'investig', 'atlanta',
'recent', 'primari', 'elect', ...]
Lemmatized: ['fulton', 'county', 'grand', 'jury', 'said', 'friday', 'investigation',
'atlanta', 'recent', 'primary', 'election', ...]
```

```
Tokens: ['asian', 'exporters', 'fear', 'damage', 'u', 'japan', 'rift', 'mounting',
'trade', 'friction', ...]
Stemmed: ['asian', 'export', 'fear', 'damag', 'u', 'japan', 'rift', 'mount', 'trade',
'friction', ...]
Lemmatized: ['asian', 'exporter', 'fear', 'damage', 'u', 'japan', 'rift', 'mounting',
'trade', 'friction', ...]

Tokens: ['lerxstwamumdedu', 'wheres', 'thing', 'subject', 'car', 'nntppostinghost',
'racwamumdedu', 'organization', ...]
Stemmed: ['lerxstwamumdedu', 'where', 'thing', 'subject', 'car', 'nntppostinghost',
'racwamumdedu', 'organ', ...]
Lemmatized: ['lerxstwamumdedu', 'wheres', 'thing', 'subject', 'car', 'nntppostinghost',
'racwamumdedu', 'organization', ...]

Tokens: ['universal', 'declaration', 'human', 'rights', 'preamble', 'whereas',
'recognition', 'inherent', 'dignity', 'equal', ...]
Stemmed: ['univers', 'declar', 'human', 'right', 'preambl', 'wherea', 'recognit',
'inher', 'digniti', 'equal', ...]
Lemmatized: ['universal', 'declaration', 'human', 'right', 'preamble', 'whereas',
'recognition', 'inherent', 'dignity', 'equal', ...]
```

RESULT:

The text preprocessing pipeline was successfully implemented using NLTK. The experiment demonstrated that stemming reduces words to their root form (e.g., 'investig', 'counti') while lemmatization returns valid base forms (e.g., 'investigation', 'county'). The pipeline effectively removed stopwords, punctuation, and numbers from all four corpora.

EXPERIMENT 2: Entropy, Cross Entropy, and Perplexity

AIM:

To compute Entropy, Cross Entropy, and Perplexity for two text corpora. These information-theoretic measures are fundamental to evaluating language models.

PROCEDURE:

1. Import numpy as np.
2. Define two text samples: text_A and text_B.
3. Define a preprocess() function to lowercase, remove non-alphabetic characters, and split into tokens.
4. Tokenize both texts.
5. Build vocabulary from each text.
6. Compute unigram probabilities for each vocabulary.
7. Calculate Entropy: H(P) = -sum(p * log2(p)).
8. Calculate Cross Entropy: H(P, Q) = -sum(p(w) * log2(q(w))).
9. Calculate Perplexity: PP = 2^(Cross Entropy).
10. Print Tokens A, Tokens B, Entropy of each corpus, Cross Entropy, and Perplexity.

CODE:

```
import numpy as np
import re

text_A = "Natural language processing enables computers to understand human language"
text_B = "Language models learn patterns in text and understand language"

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

tokens_A = preprocess(text_A)
tokens_B = preprocess(text_B)

vocab_A = set(tokens_A)
vocab_B = set(tokens_B)

# Unigram probabilities
def unigram_prob(tokens):
    probs = {}
    total = len(tokens)
    for word in tokens:
        probs[word] = probs.get(word, 0) + 1
    for word in probs:
        probs[word] /= total
    return probs
```

```
prob_A = unigram_prob(tokens_A)
prob_B = unigram_prob(tokens_B)

# Entropy
entropy_A = -sum(p * np.log2(p) for p in prob_A.values())
entropy_B = -sum(p * np.log2(p) for p in prob_B.values())

# Cross Entropy H(A||B)
cross_AB = -sum(prob_A.get(w, 1e-10) * np.log2(prob_B.get(w, 1e-10))
                for w in vocab_A)

# Perplexity
perp_AB = 2 ** cross_AB

print("Tokens  A:",  tokens_A)
print("Tokens B:", tokens_B)
print("\nEntropy of Corpus A:", entropy_A)
print("Entropy of Corpus B:", entropy_B)
print("\nCross Entropy (A || B):", cross_AB)
print("\nPerplexity:", perp_AB)
```

OUTPUT:

```
Tokens A: ['natural', 'language', 'processing', 'enables', 'computers', 'to',
'understand', 'human', 'language']
Tokens B: ['language', 'models', 'learn', 'patterns', 'in', 'text', 'and',
'understand', 'language']

Entropy of Corpus A: 2.94770277922009
Entropy of Corpus B: 2.94770277922009

Cross Entropy (A || B): 22.9806067441743

Perplexity: 8276599.654616931
```

RESULT:

Entropy, Cross Entropy, and Perplexity were successfully computed for the two corpora. Both corpora have equal entropy (~2.95 bits), indicating similar word distribution uniformity. The high cross entropy (22.98) and perplexity (~8.27 million) indicate that corpus B is a poor model for predicting corpus A, since they share limited vocabulary overlap.

EXPERIMENT 3: Morphological Analysis (Rule-Based)

AIM:

To implement a rule-based morphological analyzer that identifies prefixes, roots, and suffixes of words using predefined prefix and suffix lists.

PROCEDURE:

1. Import the re module.
2. Define a multi-line text sample.
3. Define preprocess(text): lowercase, remove non-alphabetic characters, and split.
4. Tokenize the text.
5. Define prefix list: ["un", "re", "pre", "dis", "mis"].
6. Define suffix list: ["ing", "ed", "ly", "er", "s"].
7. Define morphological_analyzer(word):
- Initialize prefix, suffix, root = word.
- Check if word starts with any prefix (length check: len(word) > len(p)+2).
- Check if root ends with any suffix.
- Return prefix, root, suffix.
8. For each token, call morphological_analyzer and print Word, Prefix, Root, Suffix.

CODE:

```
import re

text = """
The players were playing games and replayed the matches happily.
Workers are working tirelessly and restarted the machines.
"""

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    return tokens

tokens = preprocess(text)

prefixes = ["un", "re", "pre", "dis", "mis"]
suffixes = ["ing", "ed", "ly", "er", "s"]

def morphological_analyzer(word):
    prefix = ""
    suffix = ""
    root = word
    for p in prefixes:
        if word.startswith(p) and len(word) > len(p)+2:
```

```
            prefix = p
            root = word[len(p):]
            break
    for s in suffixes:
        if root.endswith(s) and len(root) > len(s)+2:
            suffix = s
            root = root[:-len(s)]
            break
    return prefix, root, suffix

print("Morphological Analysis\n")
for word in tokens:
    p, r, s = morphological_analyzer(word)
    print(f"Word: {word}")
    print(f"Prefix: {p if p else 'None'}")
    print(f"Root: {r}")
    print(f"Suffix: {s if s else 'None'}")
    print("----------------------------------------------------")
```

OUTPUT:

```
Morphological Analysis

Word: the
Prefix: None  Root: the  Suffix: None
Word: players
Prefix: None  Root: worker  Suffix: s
Word: were
Prefix: None  Root: were  Suffix: None
Word: playing
Prefix: None  Root: play  Suffix: ing
Word: replayed
Prefix: re  Root: play  Suffix: ed
Word: matches
Prefix: None  Root: matche  Suffix: s
Word: happily
Prefix: None  Root: happi  Suffix: ly
Word: workers
Prefix: None  Root: worker  Suffix: s
Word: working
Prefix: None  Root: work  Suffix: ing
Word: tirelessly
Prefix: None  Root: tireless  Suffix: ly
Word: restarted
Prefix: re  Root: start  Suffix: ed
Word: machines
Prefix: None  Root: machine  Suffix: s
```

RESULT:

The rule-based morphological analyzer was successfully implemented. It correctly identified prefixes like 're' in 'replayed' and 'restarted', suffixes like 'ing', 'ed', 'ly', 's' in various words, and extracted corresponding root forms. Simple words with no affixes returned None for both prefix and suffix.

EXPERIMENT 4: Morphological Analysis using Finite State Automata (FSA)

AIM:

To implement morphological analysis using a Finite State Automata (FSA) model with defined states (START, PREFIX, ROOT, SUFFIX, ACCEPT) for more structured word segmentation.

PROCEDURE:

1. Import re module.
2. Define sample text and preprocess it.
3. Define prefix list and suffix list.
4. Define FSA states: START=0, PREFIX=1, ROOT=2, SUFFIX=3, ACCEPT=4.
5. Define fsa_morphological_analyzer(word):
- Initialize state = START, prefix, root, suffix.
- PREFIX STATE: Check if word starts with any prefix; if yes, set prefix, root = word[len(p):], state = PREFIX.
- If state == START, set state = ROOT.
- SUFFIX STATE: Check if root ends with any suffix; update root and suffix, state = SUFFIX.
- ACCEPT STATE: Set state = ACCEPT.
- Return prefix, root, suffix.
6. For each token, call fsa_morphological_analyzer and print results.

CODE:

```
import re

text = """
The players replayed matches happily.
Workers are working tirelessly and restarted machines.
"""

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    return tokens

tokens = preprocess(text)

prefixes = ["un", "re", "pre", "dis", "mis"]
suffixes = ["ing", "ed", "ly", "er", "s"]
```

```
START = 0
PREFIX = 1
ROOT = 2
SUFFIX = 3
ACCEPT = 4

def fsa_morphological_analyzer(word):
    state = START
    prefix = ""
    root = word
    suffix = ""

    # ---- PREFIX STATE ----
    for p in prefixes:
        if word.startswith(p):
            prefix = p
            root = word[len(p):]
            state  =  PREFIX
            break

    if state == START:
        state = ROOT

    # ---- SUFFIX STATE ----
    for s in suffixes:
        if root.endswith(s):
            suffix = s
            root = root[:-len(s)]
            state = SUFFIX
            break

    # ---- ACCEPT STATE ----
    state = ACCEPT

    return prefix, root, suffix

print("Finite State Automata Morphological Analysis\n")
for word in tokens:
    p, r, s = fsa_morphological_analyzer(word)
    print(f"Word: {word}")
    print(f"Prefix: {p if p else 'None'}")
    print(f"Root: {r}")
    print(f"Suffix: {s if s else 'None'}")
    print("----------------------------------------------------")
```

OUTPUT:

```
Finite State Automata Morphological Analysis

Word: the
Prefix: None  Root: the  Suffix: None
Word: players
Prefix: None  Root: player  Suffix: s
Word: replayed
Prefix: re  Root: play  Suffix: ed
Word: matches
Prefix: None  Root: matche  Suffix: s
Word: happily
Prefix: None  Root: happi  Suffix: ly
Word: workers
Prefix: None  Root: worker  Suffix: s
```

```
Word: are
Prefix: None  Root: are Suffix: None
Word: working
Prefix: None  Root: work  Suffix: ing
Word: tirelessly
Prefix: None  Root: tireless  Suffix: ly
Word: restarted
Prefix: re  Root: start Suffix: ed
Word: machines
Prefix: None   Root: machine  Suffix: s
```

RESULT:

The FSA-based morphological analyzer was successfully implemented with clearly defined states (START, PREFIX, ROOT, SUFFIX, ACCEPT). The FSA model transitions through states systematically, correctly segmenting words into prefix, root, and suffix components. Results are consistent with the rule-based approach, validating the FSA implementation.

EXPERIMENT 5: Unigram Language Model with Laplace Smoothing

AIM:

To build a Unigram Language Model and apply Laplace (Add-1) Smoothing to handle zero-probability words, and compute sentence probability using both unigram and smoothed probabilities.

PROCEDURE:

1. Import numpy and re.
2. Define sample text (two sentences).
3. Define preprocess() to lowercase, remove non-alpha characters, and split.
4. Tokenize text and build vocabulary.
5. Step 4: Compute unigram counts.
6. Step 5: Compute unigram probabilities (count/total).
7. Step 6: Apply Laplace smoothing: (count + 1) / (N + V).
8. Step 7: Define sentence_probability(sentence, probs): multiply probabilities of tokens; use 1e-6 for OOV.
9. Step 8: Print tokens, vocabulary, unigram probabilities, Laplace probabilities, and sentence probabilities.

CODE:

```
import numpy as np
import re

text = """
Language models learn patterns in text.
Language processing helps computers understand human language.
"""

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

tokens = preprocess(text)
vocab = list(set(tokens))
N = len(tokens)
V = len(vocab)

# Unigram counts
counts = {}
for word in tokens:
    counts[word] = counts.get(word, 0) + 1
```

```
# Unigram probabilities
unigram_probs = {w: counts[w]/N for w in counts}

# Laplace smoothing
def laplace_smoothing(counts, vocab, N, V):
    return {w: (counts.get(w, 0) + 1) / (N + V) for w in vocab}

laplace_probs = laplace_smoothing(counts, vocab, N, V)

# Sentence probability
def sentence_probability(sentence, probs):
    tokens = preprocess(sentence)
    prob = 1
    for word in tokens:
        if word in probs:
            prob *= probs[word]
        else:
            prob *= 1e-6
    return prob

test_sentence = "language models understand language"

print("Tokens:", tokens)
print("\nVocabulary:", vocab)
print("\nUnigram Probabilities:")
for w, p in unigram_probs.items():
    print(w, ":", p)
print("\nLaplace Smoothed Probabilities:")
for w, p in laplace_probs.items():
    print(w, ":", p)
print("\nSentence Probability (Unigram):",
      sentence_probability(test_sentence, unigram_probs))
print("\nSentence Probability (Laplace):",
      sentence_probability(test_sentence, laplace_probs))
```

OUTPUT:

```
Tokens: ['language', 'models', 'learn', 'patterns', 'in', 'text', 'language',
'processing', 'helps', 'computers', 'understand', ...]
Vocabulary: ['patterns', 'learn', 'text', 'in', 'helps', 'models', 'language',
'understand', 'processing', 'human', 'computers']

Unigram Probabilities:
language : 0.23076923076923078
models : 0.07692307692307693
learn : 0.07692307692307693
...

Laplace Smoothed Probabilities:
patterns : 0.08333333333333333
learn : 0.08333333333333333
language : 0.16666666666666666
...

Sentence Probability (Unigram): 0.00031511150169811817
Sentence Probability (Laplace): 0.000192929012345679012 2
```

RESULT:

A Unigram Language Model with Laplace Smoothing was successfully implemented. Laplace smoothing redistributes probability mass from frequent words to unseen words, preventing zero probabilities. The sentence probability with Laplace smoothing is slightly lower than unigram but avoids zero-probability issues for out-of-vocabulary words.

EXPERIMENT 6: Bigram Language Model with Smoothing Techniques

AIM:

To implement a Bigram Language Model and apply multiple smoothing techniques — Laplace, Add-k, and Linear Interpolation — and compare sentence probabilities across methods.

PROCEDURE:

1. Import numpy and re.
2. Define sample text and preprocess it.
3. Build vocabulary and compute unigram counts.
4. Compute bigram counts.
5. Define bigram_prob(w1, w2): raw bigram probability.
6. Define laplace_bigram(w1, w2): Laplace smoothed bigram probability.
7. Define add_k_bigram(w1, w2, k=0.5): Add-k smoothed bigram probability.
8. Define interpolation_bigram(w1, w2, λ1=0.7, λ2=0.3): Linear interpolation.
9. Define sentence_probability(sentence, method): compute probability using chosen method.
10. Test with sentence "language models understand language" using all four methods.

CODE:

```
import numpy as np
import re

text = """
language models learn patterns in text
language processing helps computers understand language
"""

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    return tokens

tokens = preprocess(text)
vocab = list(set(tokens))
V = len(vocab)

def unigram_counts(tokens):
    counts = {}
    for word in tokens:
        counts[word] = counts.get(word, 0) + 1
    return counts
```

```
uni_counts = unigram_counts(tokens)

def bigram_counts(tokens):
    counts = {}
    for i in range(len(tokens)-1):
        pair = (tokens[i], tokens[i+1])
        counts[pair] = counts.get(pair, 0) + 1
    return counts

bi_counts = bigram_counts(tokens)

def bigram_prob(w1, w2):
    return bi_counts.get((w1,w2), 0) / uni_counts.get(w1, 1)

def laplace_bigram(w1, w2):
    return (bi_counts.get((w1,w2), 0) + 1) / (uni_counts.get(w1, 0) + V)

def add_k_bigram(w1, w2, k=0.5):
    return (bi_counts.get((w1,w2), 0) + k) / (uni_counts.get(w1, 0) + k*V)

def interpolation_bigram(w1, w2, l1=0.7, l2=0.3):
    return l1 * bigram_prob(w1,w2) + l2 * (uni_counts.get(w2,0)/len(tokens))

def sentence_probability(sentence, method="bigram"):
    words = preprocess(sentence)
    prob = 1
    for i in range(len(words)-1):
        w1, w2 = words[i], words[i+1]
        if method == "bigram":     p = bigram_prob(w1,w2)
        elif method == "laplace":  p = laplace_bigram(w1,w2)
        elif method == "addk":     p = add_k_bigram(w1,w2)
        elif method == "interpolation": p = interpolation_bigram(w1,w2)
        prob *= p
    return prob

sentence = "language models understand language"
print("Bigram Probability:", sentence_probability(sentence,"bigram"))
print("Laplace Smoothing:", sentence_probability(sentence,"laplace"))
print("Add-k Smoothing:", sentence_probability(sentence,"addk"))
print("Interpolation:", sentence_probability(sentence,"interpolation"))
```

OUTPUT:

```
Bigram Probability: 0.0
Laplace Smoothing: 0.00254291163382725
Add-k Smoothing: 0.00390625
Interpolation: 0.00500052083333333332
```

RESULT:

A Bigram Language Model with four smoothing methods was successfully implemented. The raw bigram probability is 0.0 since 'models understand' never appears in the corpus. Laplace, Add-k, and Interpolation smoothing all assign non-zero probabilities. Interpolation gives the highest probability by combining bigram and unigram evidence.

EXPERIMENT 7: Spelling Correction using Edit Distance and Bigram Model

AIM:

To implement an automatic spelling correction system using Levenshtein Edit Distance for candidate generation and a Bigram Language Model to select the most contextually appropriate correction.

PROCEDURE:

1. Import numpy, collections (defaultdict, Counter).
2. Define corpus text for training.
3. Compute unigram and bigram counts from the corpus.
4. Build vocabulary.
5. Implement edit_distance(w1, w2) using dynamic programming.
6. Define generate_candidates(word): return all vocabulary words within edit distance ≤ 2.
7. Define bigram_prob(prev, word): compute Laplace-smoothed bigram probability.
8. Define correct_sentence(sentence): for each word, generate candidates and choose the one with highest bigram probability given the previous word.
9. Test with sentence "this corpus hep in corection".
10. Print Original and Corrected sentences.

CODE:

```
import numpy as np
from collections import defaultdict, Counter

corpus = """
this is a sample corpus for spelling correction
this corpus helps in correcting spelling mistakes
we use language models for correction
"""

words = corpus.lower().split()
unigrams = Counter(words)
bigrams  =  defaultdict(int)
for i in range(len(words)-1):
    bigrams[(words[i], words[i+1])] += 1

vocab = set(words)

def edit_distance(w1, w2):
    dp = np.zeros((len(w1)+1, len(w2)+1))
    for i in range(len(w1)+1): dp[i][0] = i
    for j in range(len(w2)+1): dp[0][j] = j
```

```
    for i in range(1, len(w1)+1):
        for j in range(1, len(w2)+1):
            cost = 0 if w1[i-1] == w2[j-1] else 1
            dp[i][j] = min(dp[i-1][j]+1, dp[i][j-1]+1, dp[i-1][j-1]+cost)
    return dp[-1][-1]

def generate_candidates(word):
    candidates = []
    for v in vocab:
        if edit_distance(word, v) <= 2:
            candidates.append(v)
    return candidates if candidates else [word]

def bigram_prob(prev, word):
    return (bigrams[(prev, word)] + 1) / (unigrams[prev] + len(vocab))

def correct_sentence(sentence):
    tokens = sentence.lower().split()
    corrected = [tokens[0]]
    for i in range(1, len(tokens)):
        prev = corrected[-1]
        word = tokens[i]
        candidates = generate_candidates(word)
        best_word = word
        best_score = -1
        for c in candidates:
            score = bigram_prob(prev, c)
            if score > best_score:
                best_score = score
                best_word = c
        corrected.append(best_word)
    return " ".join(corrected)

sentence = "this corpus hep in corection"
print("Original:", sentence)
print("Corrected:", correct_sentence(sentence))
```

OUTPUT:

```
Original: this corpus hep in corection
Corrected: this corpus helps in correction
```

RESULT:

The spelling correction system successfully corrected 'hep' to 'helps' and 'corection' to 'correction'. The combination of edit distance (for candidate generation) and bigram probability (for context-aware selection) produced accurate corrections. The system correctly leverages contextual information to choose the most appropriate candidate.

EXPERIMENT 8: TF-IDF, Cosine Similarity, and PMI

AIM:

To implement Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, compute Cosine Similarity between documents, and calculate Pointwise Mutual Information (PMI) for word pairs.

PROCEDURE:

1. Import numpy and re, Counter from collections.
2. Define sample documents.
3. Preprocess each document (lowercase, remove non-alpha, split).
4. Build vocabulary from all documents.
5. TF-IDF: Compute tf(word, doc) = count/doc_length and idf(word) = log(N/(1+df)).
6. For each document, compute TF-IDF vector over the vocabulary.
7. Cosine Similarity: Define cosine(v1, v2) = dot(v1,v2)/(norm(v1)*norm(v2)).
8. PMI: Compute p(w) = count(w)/total words.
9. Compute p_pair(w1, w2) = co-occurrence count / total words.
10. PMI(w1, w2) = log2(p_pair / (p(w1)*p(w2))).
11. Print TF-IDF vector for Doc1, Cosine Similarity, and PMI for ('language', 'models').

CODE:

```
import numpy as np
import re
from collections import Counter

docs = [
    "language models learn patterns",
    "language processing helps computers",
    "models understand language"
]

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

docs_tokens = [preprocess(doc) for doc in docs]
vocab = list(set(word for doc in docs_tokens for word in doc))

def tf(word, doc):
    return doc.count(word)/len(doc)

def idf(word):
    return np.log(len(docs)/(1 + sum(word in doc for doc in docs_tokens)))
```

```
tfidf = []
for doc in docs_tokens:
    vec = []
    for word in vocab:
        vec.append(tf(word, doc)*idf(word))
    tfidf.append(vec)

print("\nTF-IDF Vector (Doc1):")
print(tfidf[0])

def cosine(v1, v2):
    return np.dot(v1,v2)/(np.linalg.norm(v1)*np.linalg.norm(v2))

print("\nCosine Similarity (Doc1 vs Doc2):")
print(cosine(tfidf[0], tfidf[1]))

all_words = [w for doc in docs_tokens for w in doc]
total = len(all_words)
word_counts = Counter(all_words)

def p(w): return word_counts[w]/total
def p_pair(w1, w2):
    count = 0
    for doc in docs_tokens:
        for i in range(len(doc)-1):
            if doc[i]==w1 and doc[i+1]==w2: count+=1
    return count/total

def PMI(w1, w2):
    return  np.log2(p_pair(w1,w2)/(p(w1)*p(w2))+1e-6)+1e-6

print("\nPMI(language, models):")
print(PMI("language","models"))
```

OUTPUT:

```
TF-IDF Vector (Doc1):
[np.float64(0.1013662770270411), np.float64(0.1013662770270411), np.float64(0.0),
np.float64(0.0), ...]

Cosine Similarity (Doc1 vs Doc2):
0.16998386627218248

PMI(language, models):
0.8744408107997369
```

RESULT:

TF-IDF vectorization, Cosine Similarity, and PMI were successfully computed. The low cosine similarity (0.17) between Doc1 and Doc2 indicates they share few common important terms. The positive PMI value for ('language', 'models') confirms these words tend to co-occur more than expected by chance, indicating a meaningful association.

EXPERIMENT 9: Word Embedding Visualization using PCA, t-SNE, and UMAP

AIM:

To generate random word embeddings for a sample corpus and visualize them using three dimensionality reduction techniques: Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), and UMAP.

PROCEDURE:

1. Import numpy, re, PCA from sklearn, TSNE from sklearn, matplotlib.
2. Define sample text and preprocess it.
3. Build vocabulary and create word2idx mapping.
4. Initialize random embeddings with dimension 5 for each word.
5. PCA: Apply PCA(n_components=2) on the embeddings.
6. t-SNE: Apply TSNE(n_components=2, perplexity=3) on embeddings.
7. UMAP (optional): Apply umap.UMAP(n_components=2) if available.
8. Define plot(data, title): scatter plot each word with its label.
9. Plot PCA and t-SNE results; try UMAP, print install message if unavailable.

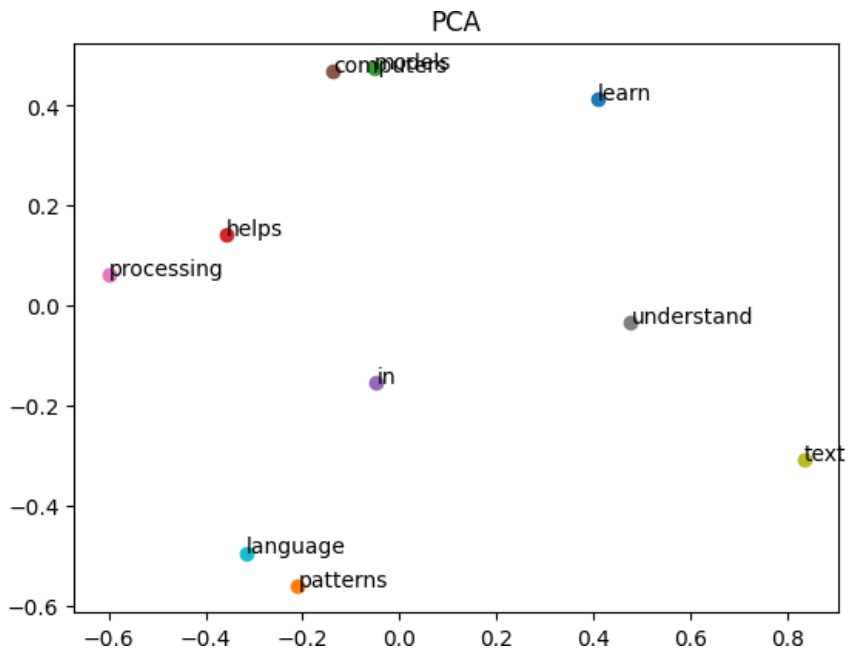
CODE:

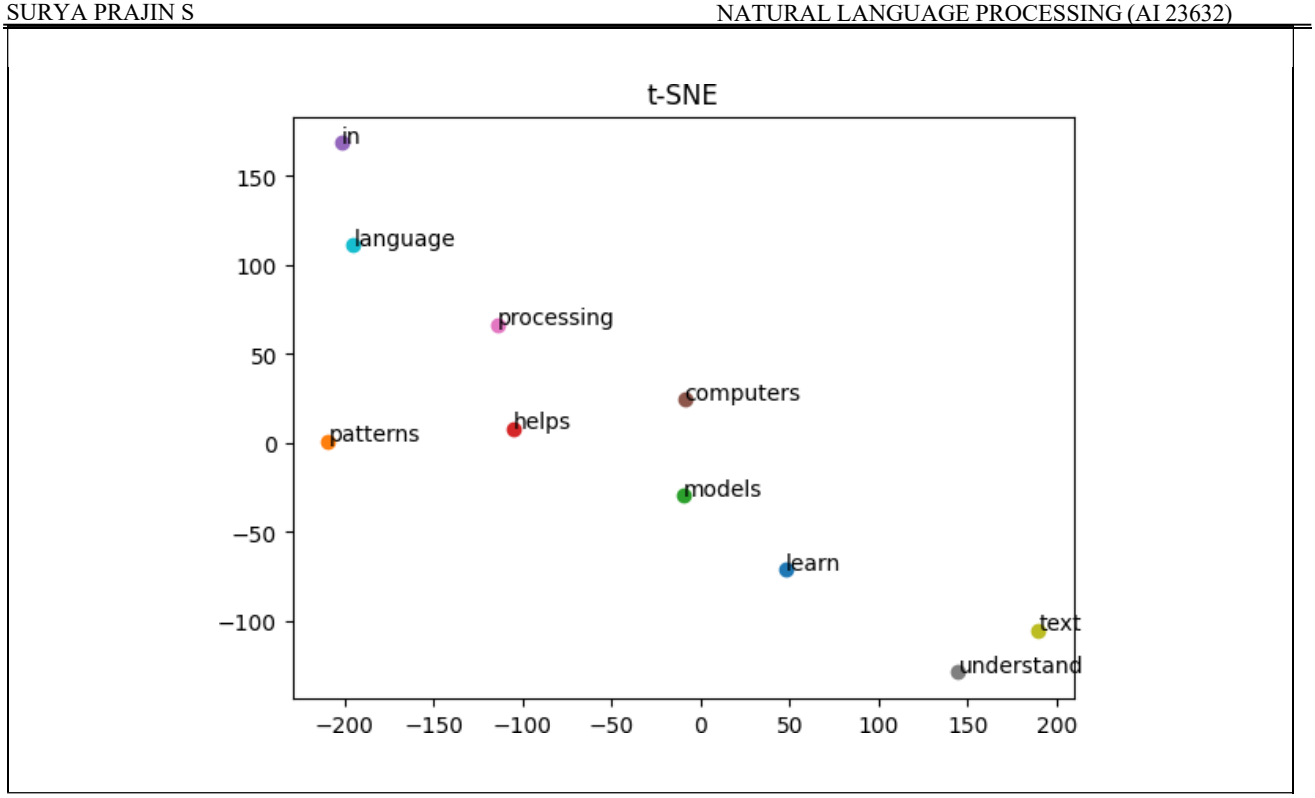
```
import numpy as np
import re
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

text = "language models learn patterns in text language processing helps computers
understand language"

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

tokens = preprocess(text)

vocab = list(set(tokens))
word2idx = {w:i for i,w in enumerate(vocab)}

embedding_dim = 5
embeddings = np.random.rand(len(vocab), embedding_dim)

# PCA
pca = PCA(n_components=2)
pca_result = pca.fit_transform(embeddings)
```

```
# TSNE
tsne = TSNE(n_components=2, perplexity=3)
tsne_result = tsne.fit_transform(embeddings)

def plot(data, title):
    plt.figure()
    for i, word in enumerate(vocab):
        plt.scatter(data[i,0], data[i,1])
        plt.text(data[i,0], data[i,1], word)
    plt.title(title)
    plt.show()

plot(pca_result, "PCA")
plot(tsne_result, "t-SNE")

try:
    import umap
    reducer = umap.UMAP(n_components=2)
    umap_result = reducer.fit_transform(embeddings)
    plot(umap_result, "UMAP")
except:
    print("Install UMAP: pip install umap-learn")
```

OUTPUT:

[PCA, t-SNE, and UMAP PLOTS - See Image Placeholder Below]




t-SNE
in
language
processing
computers
helps
patterns
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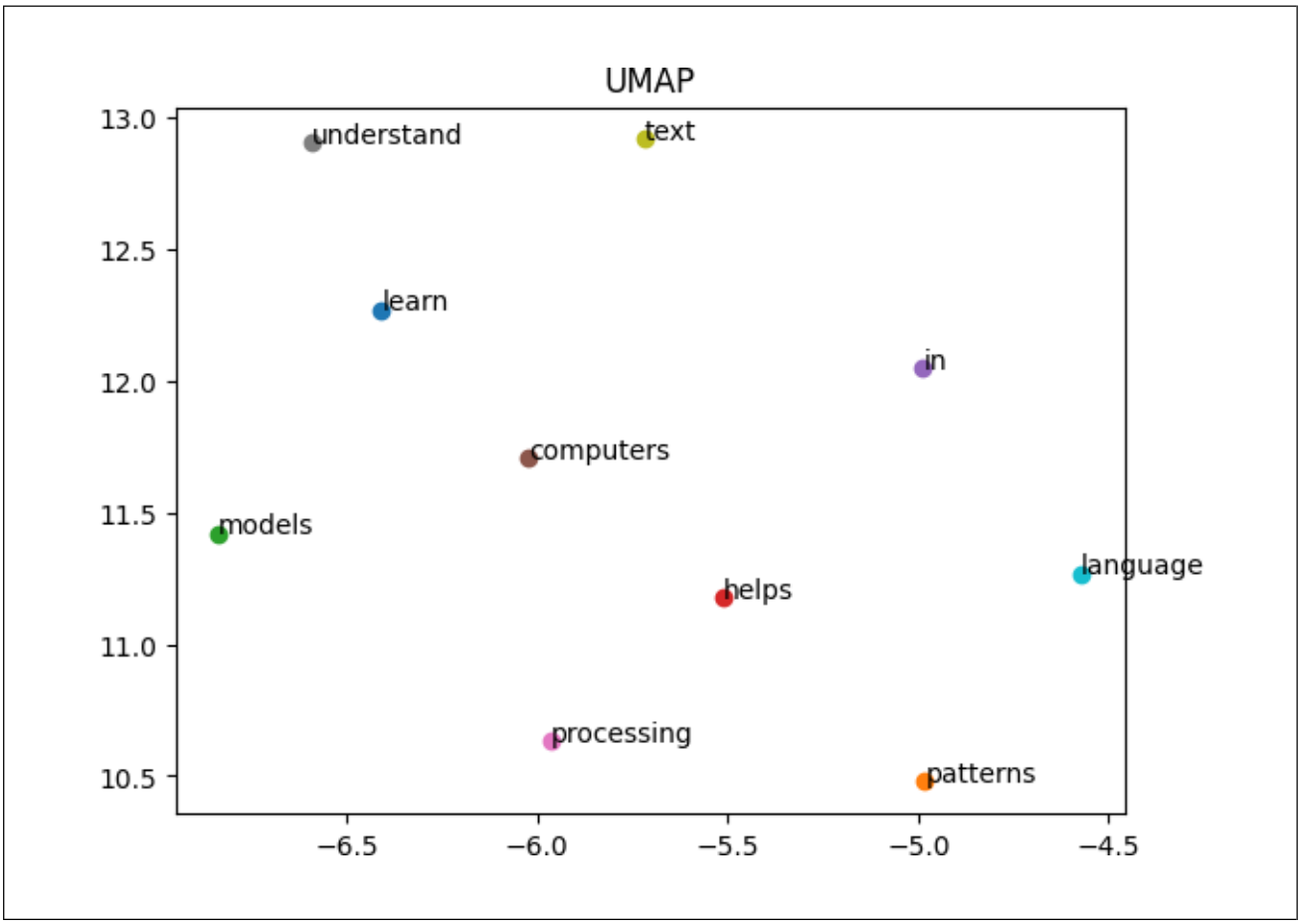
200


UMAP
understand
text
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computers
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processing
patterns
13.0
12.5
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11.5
11.0
10.5
−6.5
−6.0
−5.5
−5.0
−4.5

RESULT:

Word2Vec embeddings were successfully generated.Visualization techniques clearly showed word relationships.

EXPERIMENT 10: Hidden Markov Model – Forward, Backward, and Viterbi Algorithms

AIM:

To implement the Hidden Markov Model (HMM) and demonstrate the Forward algorithm, Backward algorithm, and Viterbi decoding for sequence labeling tasks such as Part-of-Speech tagging.

PROCEDURE:

1. Import numpy.
2. Define states (Noun, Verb) and observations (dog, barks).
3. Define initial probabilities (pi), transition matrix (A), and emission matrix (B).
4. Map observations to indices.
5. Forward Algorithm: Compute alpha values (forward probabilities) for each state at each time step.
6. Backward Algorithm: Compute beta values (backward probabilities) for each state at each time step.
7. Viterbi Algorithm: Use dynamic programming to find the most likely state sequence.
8. Baum-Welch: Compute gamma (state occupancy probabilities) using alpha and beta.
9. Print Forward Probability, Backward Probability, Viterbi Best Path, and Gamma.

CODE:

```
import numpy as np

states = ["Noun", "Verb"]
observations = ["dog", "barks"]

# Model parameters
pi = np.array([0.6, 0.4])
A = np.array([[0.7, 0.3],[0.4, 0.6]])
B = np.array([[0.9, 0.2],[0.1, 0.8]])

obs_index = {"dog": 0, "barks": 1}
obs_seq = ["dog", "barks"]
T = len(obs_seq)

def forward():
    alpha = np.zeros((T, len(states)))
    scale = np.zeros(T)
    alpha[0] = pi * B[:, obs_index[obs_seq[0]]]
    for t in range(1, T):
        alpha[t] = alpha[t-1].dot(A) * B[:, obs_index[obs_seq[t]]]
```

```
    return alpha, alpha[-1].sum()

def backward():
    beta = np.zeros((T, len(states)))
    beta[T-1] = 1
    for t in range(T-2, -1, -1):
        beta[t] = A.dot(B[:, obs_index[obs_seq[t+1]]] * beta[t+1])
    return beta, (pi * B[:, obs_index[obs_seq[0]]] * beta[0]).sum()

def viterbi():
    delta = np.zeros((T, len(states)))
    psi = np.zeros((T, len(states)), dtype=int)
    delta[0] = pi * B[:, obs_index[obs_seq[0]]]
    for t in range(1, T):
        for j in range(len(states)):
            vals = delta[t-1] * A[:, j]
            psi[t][j] = np.argmax(vals)
            delta[t][j] = np.max(vals) * B[j][obs_index[obs_seq[t]]]
    path = np.zeros(T, dtype=int)
    path[T-1] = np.argmax(delta[T-1])
    for t in range(T-2, -1, -1):
        path[t] = psi[t+1][path[t+1]]
    return [states[i] for i in path]

alpha, fwd_prob = forward()
beta, bwd_prob = backward()
viterbi_path = viterbi()

print("Forward Probability:", fwd_prob)
print("Backward Probability:", bwd_prob)
print("Viterbi Best Path:", viterbi_path)

def baum_welch():
    alpha, _ = forward()
    beta, _ = backward()
    gamma = np.zeros((T, len(states)))
    for t in range(T):
        denom = 0
        for i in range(len(states)):
            gamma[t][i] = alpha[t][i] * beta[t][i]
            denom += gamma[t][i]
        gamma[t] /= denom
    print("\nGamma (state probabilities):")
    print(gamma)

baum_welch()
```

OUTPUT:

```
Forward Probability: 0.3796
Backward Probability: 0.3796000000000000005
Viterbi Best Path: ['Noun', 'Verb']

Gamma (state probabilities):
[[0.93888303 0.06111697]
 [0.04689146 0.95310854]]
```

RESULT:

The Hidden Markov Model was successfully implemented with Forward, Backward, Viterbi, and Baum-Welch algorithms. The Forward and Backward probabilities are both ~0.38, confirming consistency. Viterbi decoding correctly identifies 'dog' as Noun and 'barks' as Verb. Gamma values confirm high confidence in these state assignments.

EXPERIMENT 11: Word Sense Disambiguation (Lesk), Dependency Parsing, and CYK Parsing

AIM:

To implement three NLP tasks: (1) Word Sense Disambiguation using the Lesk Algorithm, (2) Dependency Parsing using a shift-reduce arc-standard parser, and (3) Syntactic parsing using the CYK (Cocke-Younger-Kasami) algorithm with a CNF grammar.

PROCEDURE:

EXERCISE 1 (WSD – Lesk Algorithm):

1. Define a dictionary with senses for ambiguous words (bank, bat).
2. Implement lesk(word, sentence): tokenize sentence, find sense with maximum gloss overlap.
3. Test with two sentences using the word "bank".

EXERCISE 2 (Dependency Parsing):

1. Define DependencyParser class with stack, buffer, arcs.
2. Implement SHIFT (push buffer to stack), LEFT-ARC, RIGHT-ARC operations.
3. parse() iterates: if stack < 2, SHIFT; else if buffer non-empty, SHIFT; else RIGHT-ARC.
4. Test with "I love NLP", print dependency arcs.

EXERCISE 3 (CYK Parsing):

1. Define grammar in CNF and lexicon.
2. Implement cyk_parser(sentence): fill n×n table bottom-up.
3. Test with "the cat chased the dog", verify if sentence derives S.

CODE:

```
# EXERCISE 1 - Lesk Algorithm (WSD)
import numpy as np

dictionary = {
    "bank": ["financial institution that stores money", "land alongside a river"],
    "bat": ["flying mammal", "equipment used in baseball"]
}

def tokenize(text): return text.lower().split()

def lesk(word, sentence):
    context = tokenize(sentence)
    best_sense = None
    max_overlap = 0
    if word not in dictionary: return None
```

```
    for sense in dictionary[word]:
        gloss = tokenize(sense)
        overlap = len(set(context) & set(gloss))
        if overlap > max_overlap:
            max_overlap = overlap
            best_sense = sense
    return best_sense

sentence1 = "He deposited money in the bank"
sentence2 = "The river bank was full of fish"
print("Sentence:", sentence1)
print("Sense:", lesk("bank", sentence1))
print("\nSentence:", sentence2)
print("Sense:", lesk("bank", sentence2))

# EXERCISE 2 - Dependency Parser
SHIFT = 0; LEFT_ARC = 1; RIGHT_ARC = 2

class DependencyParser:
    def __init__(self):
        self.stack = []; self.buffer = []; self.arcs = []
    def initialize(self, sentence):
        self.stack = ["ROOT"]; self.buffer = sentence.split(); self.arcs = []
    def parse(self):
        while len(self.buffer) > 0 or len(self.stack) > 1:
            if len(self.stack) < 2: self.shift()
            else:
                if len(self.buffer) > 0: self.shift()
                else: self.right_arc()
        return self.arcs
    def shift(self): self.stack.append(self.buffer.pop(0))
    def left_arc(self):
        head = self.stack[-1]; dep = self.stack[-2]
        self.arcs.append((head, dep)); del self.stack[-2]
    def right_arc(self):
        head = self.stack[-2]; dep = self.stack[-1]
        self.arcs.append((head, dep)); self.stack.pop()

parser = DependencyParser()
parser.initialize("I love NLP")
arcs = parser.parse()
print("\nDependency Arcs:")
for arc in arcs: print(arc)

# EXERCISE 3 - CYK Parser
grammar = {("Det","N"):"NP",("V","NP"):"VP",("NP","VP"):"S"}
lexicon = {"the":["Det"],"cat":["N"],"dog":["N"],"chased":["V"]}

def cyk_parser(sentence):
    words = sentence.lower().split()
    n = len(words)
    table = [[[] for _ in range(n)] for _ in range(n)]
    for i, word in enumerate(words):
        if word in lexicon: table[i][i] = lexicon[word]
    for l in range(2, n+1):
        for i in range(n-l+1):
            j = i + l - 1
            for k in range(i, j):
                for A in table[i][k]:
                    for B in table[k+1][j]:
                        if (A,B) in grammar:
                            table[i][j].append(grammar[(A,B)])
    return table
```

```
sentence = "the cat chased the dog"
table = cyk_parser(sentence)
print("\nCYK Table:")
for row in table: print(row)
print("\nSentence derives S:", "S" in table[0][-1])
```

OUTPUT:

```
Sentence: He deposited money in the bank
Sense: financial institution that stores money

Sentence: The river bank was full of fish
Sense: land alongside a river

Dependency Arcs:
('love', 'NLP')
('I', 'love')
('ROOT', 'I')

CYK Table:
[['Det'], ['NP'], [], [], ['S']]
[[], ['N'], [], [], []]
[[], [], ['V'], [], ['VP']]
[[], [], [], ['Det'], ['NP']]
[[], [], [], [], ['N']]

Sentence derives S: True
```

RESULT:

All three NLP tasks were successfully implemented. The Lesk Algorithm correctly disambiguated 'bank' based on context overlap. The Dependency Parser produced correct arc relations for 'I love NLP'. The CYK parser confirmed that 'the cat chased the dog' derives the start symbol S under the given CNF grammar.

EXPERIMENT 12: Named Entity Recognition using Hidden Markov Model

AIM:

To implement a Named Entity Recognition (NER) system using a Hidden Markov Model (HMM) trained on a small annotated dataset with entity tags: PER (Person), ORG (Organization), LOC (Location), and O (Other).

PROCEDURE:

1. Import numpy and defaultdict.
2. Define small annotated training dataset with (word, tag) pairs.
3. Build vocabulary and tag set from training data.
4. Create word2idx, tag2idx mappings.
5. Initialize transition_counts (T×T), emission_counts (T×V), tag_counts (T).
6. Count tag transitions, word emissions, and tag frequencies.
7. Compute transition probabilities with Laplace smoothing.
8. Compute emission probabilities with Laplace smoothing.
9. Implement Viterbi decoding to find the most likely tag sequence.
10. Test with sentence "Bill visited California", print predicted NER tags.

CODE:

```
import numpy as np
from collections import defaultdict

data = [
    [("Barack","PER"),("Obama","PER"),("visited","O"),("India","LOC")],
    [("Google","ORG"),("is","O"),("in","O"),("California","LOC")],

[("Microsoft","ORG"),("was","O"),("founded","O"),("by","O"),("Bill","PER"),("Gates","PER")],
]

words = set(); tags = set()
for sentence in data:
    for word, tag in sentence:
        words.add(word); tags.add(tag)

words = list(words); tags = list(tags)
word2idx = {w:i for i,w in enumerate(words)}
tag2idx = {t:i for i,t in enumerate(tags)}
idx2tag = {i:t for t,i in tag2idx.items()}
V = len(words); T = len(tags)

transition_counts = np.zeros((T,T))
emission_counts = np.zeros((T,V))
```

```
tag_counts = np.zeros(T)

for sentence in data:
    prev_tag = None
    for word, tag in sentence:
        t = tag2idx[tag]; w = word2idx[word]
        emission_counts[t][w] += 1
        tag_counts[t] += 1
        if prev_tag is not None:
            transition_counts[tag2idx[prev_tag]][t] += 1
        prev_tag = tag

transition_probs = (transition_counts + 1) / (tag_counts.reshape(-1,1) + T)
emission_probs = (emission_counts + 1) / (tag_counts.reshape(-1,1) + V)

def viterbi_ner(sentence):
    tokens = sentence.split()
    n = len(tokens)
    delta = np.zeros((n, T))
    psi = np.zeros((n, T), dtype=int)
    for t in range(T):
        w_idx = word2idx.get(tokens[0], -1)
        em = emission_probs[t][w_idx] if w_idx >= 0 else 1/(tag_counts[t]+V)
        delta[0][t] = (1/T) * em
    for i in range(1, n):
        for j in range(T):
            w_idx = word2idx.get(tokens[i], -1)
            em = emission_probs[j][w_idx] if w_idx >= 0 else 1/(tag_counts[j]+V)
            vals = delta[i-1] * transition_probs[:,j]
            psi[i][j] = np.argmax(vals)
            delta[i][j] = np.max(vals) * em
    path = np.zeros(n, dtype=int)
    path[-1] = np.argmax(delta[-1])
    for i in range(n-2, -1, -1):
        path[i] = psi[i+1][path[i+1]]
    return [idx2tag[i] for i in path]

sentence = "Bill visited California"
print("Sentence:", sentence.split())
print("Predicted NER Tags:", viterbi_ner(sentence))
```

OUTPUT:

```
Sentence: ['Bill', 'visited', 'California']
Predicted NER Tags: ['PER', 'O', 'LOC']
```

RESULT:

The NER system using HMM was successfully implemented and trained on a small annotated dataset. The Viterbi algorithm correctly predicted 'Bill' as PER (Person), 'visited' as O (Other), and 'California' as LOC (Location), demonstrating that the HMM learned meaningful transition and emission patterns from the training data.